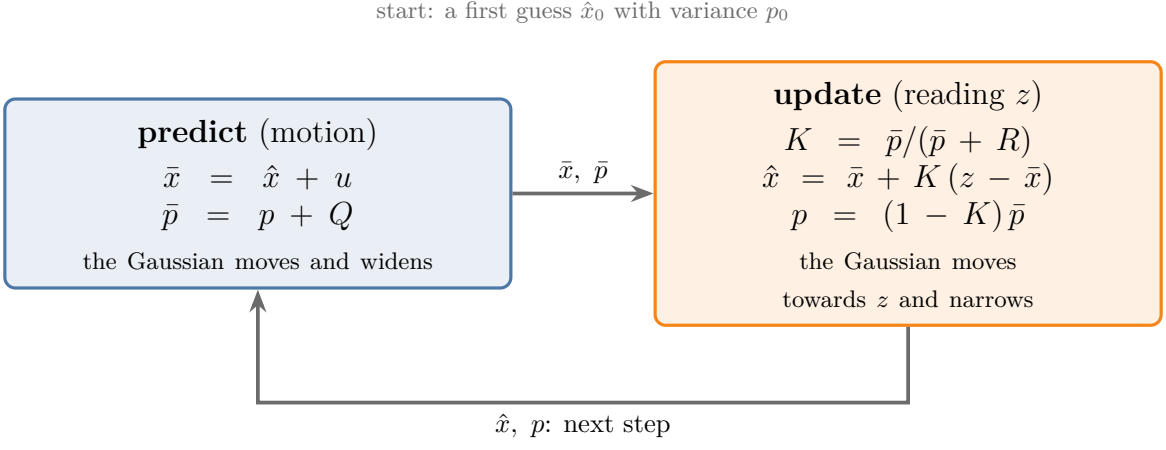

start: a first guess $\hat{x}_0$ with variance p_0
predict (motion)
$\bar{x} = \hat{x} + u$
$\bar{p} = p + Q$
the Gaussian moves and widens
$\bar{x},\ \bar{p}$
update (reading z)
$K = \bar{p}/(\bar{p} + R)$
$\hat{x} = \bar{x} + K\,(z - \bar{x})$
$p = (1 - K)\,\bar{p}$
the Gaussian moves
towards z and narrows
$\hat{x},\ p$: next step